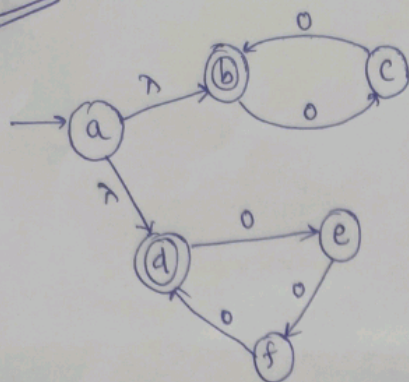


Q.5

NFA- λ to FA



1



$$\Rightarrow \epsilon(a) = \{b, d\}$$

$$\epsilon(b) = \{b\}$$

$$\epsilon(c) = \{c\}$$

$$\epsilon(d) = \{d\}$$

$$\epsilon(e) = \{e\}$$

$$\epsilon(f) = \{f\}$$

$$\Rightarrow \delta'(a, 0) = \epsilon(\delta(\epsilon(a), 0))$$

$$= \epsilon(c, e)$$

$$= \{c, e\}$$

$$\delta'(a, 1) = \epsilon(\delta(\epsilon(a), 1))$$

$$= \epsilon(\delta(\{a, b, d\}, 1))$$

$$= \epsilon(\emptyset)$$

$$= \emptyset$$

$$\Rightarrow \delta'(b, 0) = \epsilon(\delta(\epsilon(b), 0))$$

$$= \epsilon(\delta(b, 0))$$

$$= \epsilon(c)$$

$$= c$$

$$\Rightarrow \delta'(c, 0) = \epsilon(\delta(\epsilon(c), 0))$$

$$= \epsilon(\delta(c, 0))$$

$$= \epsilon(b)$$

$$= b$$

$$\Rightarrow \delta'(d, 0) = \epsilon(\delta(\epsilon(d), 0))$$

$$= \epsilon(\delta(d, 0))$$

$$= \epsilon(e)$$

$$= e$$

$$\Rightarrow \delta'(e, 0) = \epsilon(\delta(\epsilon(e), 0))$$

$$= \epsilon(\delta(e, 0))$$

$$= \epsilon(f)$$

$$= f$$

$$\Rightarrow \delta'(f, 0) = \epsilon(\delta(\epsilon(f), 0))$$

$$= \epsilon(\delta(f, 0))$$

$$= \epsilon(d)$$

$$= d$$

$$\Rightarrow \delta'(b, 1) = \emptyset$$

$$\delta'(c, 1) = \emptyset$$

$$\delta'(d, 1) = \emptyset$$

$$\delta'(e, 1) = \emptyset$$

$$\delta'(f, 1) = \emptyset$$

* transition table of NFA

δ	0
$\rightarrow a$	$\{c, e\}$
(b)	$\{c\}$
c	$\{b\}$
(d)	$\{e\}$
e	$\{f\}$
f	$\{d\}$

(2)

* NFA to DFA

from NFA table,

$\delta'(a, 0) = \{c, e\} \rightarrow$ New state

$\delta'(b, 0) = \{c\}$

$\delta'(c, 0) = \{b\}$

$\delta'(d, 0) = \{e\}$

$\delta'(e, 0) = \{f\}$

$\delta'(f, 0) = \{d\}$

$\delta'((c, e), 0) = \{b, f\} \rightarrow$ New state

$\delta'((b, f), 0) = \{c, d\} \rightarrow$ New state

$\delta'((c, d), 0) = \{b, e\} \rightarrow$ New state

$\delta'((b, e), 0) = \{c, f\} \rightarrow$ New state

$\delta'((c, f), 0) = \{b, d\} \rightarrow$ New state

$\delta'((b, d), 0) = \{c, e\} \rightarrow$ New state

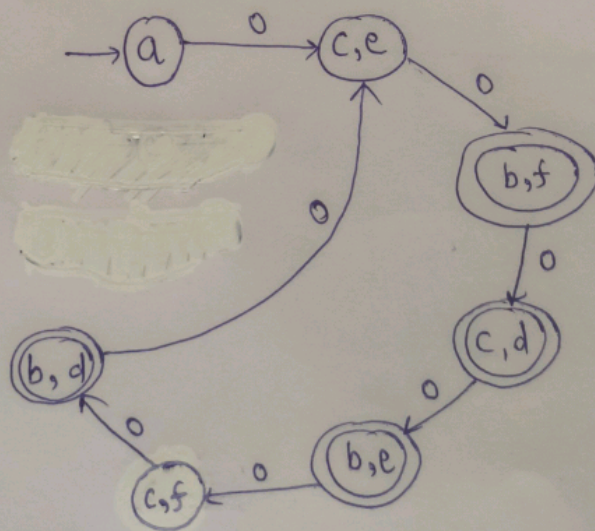
$\delta'((c, e), 0) = \{b, f\}$

* transition table of above transitions.

δ	0
$\rightarrow a$	$\{c, e\}$
$\{c, e\}$	$\{b, f\}$
$\{b, f\}^*$	$\{c, d\}$
$\{c, d\}^*$	$\{b, e\}$
$\{b, e\}^*$	$\{c, f\}$
$\{c, f\}$	$\{b, d\}$
$\{b, d\}^*$	$\{c, e\}$
$\{b, f\}^*$	$\{c\}$
c	$\{b\}$
$\{d\}^*$	$\{e\}$
e	$\{f\}$
f	$\{d\}$

* transition diagram of DFA

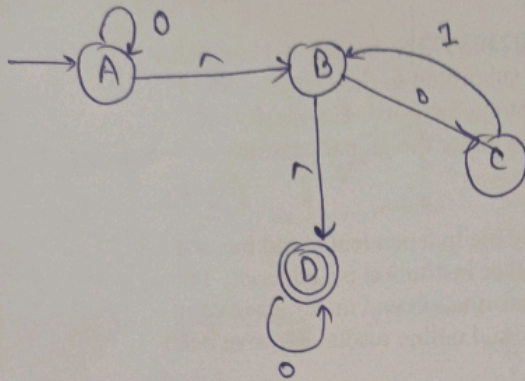
Final Answer



Q. 6 NFA $\rightarrow$ FA

(3)

transition diagram from given table:



Step 1:

$$\begin{aligned} \epsilon(A) &= \{A, B, D\} & \epsilon(C) &= \{C\} \\ \epsilon(B) &= \{B, D\} & \epsilon(D) &= \{D\} \end{aligned}$$

Step 2:

$$\begin{aligned} \delta'(A, 0) &= \epsilon(\delta(\epsilon(A), 0)) \\ &= \epsilon(\delta(\{A, B, D\}, 0)) \\ &= \epsilon(A, C, D) \\ &= \{A, B, D\} \cup \{C\} \cup \{D\} \\ &= \{A, B, C, D\} \end{aligned}$$

$$\begin{aligned} \delta'(A, 1) &= \epsilon(\delta(\epsilon(A), 1)) \\ &= \epsilon(\delta(\{A, B, D\}, 1)) \\ &= \epsilon(\emptyset) \\ &= \emptyset \end{aligned}$$

$$\begin{aligned} \delta'(B, 0) &= \epsilon(\delta(\epsilon(B), 0)) \\ &= \epsilon(\delta(\{B, D\}, 0)) \\ &= \epsilon(C, D) \\ &= \{C, D\} \end{aligned}$$

$$\begin{aligned} \delta'(B, 1) &= \epsilon(\delta(\epsilon(B), 1)) \\ &= \epsilon(\delta(\{B, D\}, 1)) \\ &= \epsilon(\emptyset) \\ &= \emptyset \end{aligned}$$

$$\delta'(C, 0) = \emptyset$$

$$\begin{aligned} \delta'(C, 1) &= \epsilon(\delta(\epsilon(C), 1)) \\ &= \epsilon(\delta(\{C\}, 1)) \\ &= \epsilon(B) \\ &= \{B, D\} \end{aligned}$$

$$\begin{aligned} \delta'(D, 0) &= \epsilon(\delta(\epsilon(D), 0)) \\ &= \epsilon(\delta(\{D\}, 0)) \\ &= \epsilon(D) \\ &= \{D\} \end{aligned}$$

$$\begin{aligned} \delta'(D, 1) &= \epsilon(\delta(\epsilon(D), 1)) = \epsilon(\delta(\{D\}, 1)) \\ &= \epsilon(\emptyset) = \emptyset \end{aligned}$$

* transition table of NFA from step 2.

(4)

input states	0	1
→ A	{A, B, C, D}	∅
B	{C, D}	∅
C	∅	{B, D}
D	{D}	∅

Step 3: NFA to DFA

$$\delta'(A, 0) = \{A, B, C, D\} \Rightarrow \text{New state}$$

$$\delta'(B, 0) = \{C, D\} \Rightarrow \text{New state}$$

$$\delta'(C, 1) = \{B, D\} \Rightarrow \text{New State}$$

$$\delta'(D, 0) = \{D\}$$

$$\delta'(\{A, B, C, D\}, 0) = \{A, B, C, D\} \quad \delta'(\{A, B, C, D\}, 1) = \{B, D\}$$

$$\delta'(\{A, B, C, D\}, 1) = \{B, D\}$$

$$\delta'(\{B, D\}, 0) = \{C, D\}$$

$$\delta'(\{B, D\}, 1) = \emptyset$$

$$\delta'(\{C, D\}, 0) = \{D\}$$

$$\delta'(\{C, D\}, 1) = \{B, D\}$$

Step 4: transition table:

	0	1
→ A	{A, B, C, D}	∅
B	{C, D}	∅
C	∅	{B, D}
D*	{D}	∅
{A, B, C, D}*	{A, B, C, D}	{B, D}
{C, D}*	{D}	{B, D}
{B, D}*	{C, D}	∅

Step 5: transition diagram

